

MODEL QUESTION PAPER

PETROLEUM REFINING & PETROCHEMICAL TECHNOLOGY

Max. Marks : 75

Part A

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. Name the major component of Natural gas
2. For crude lighter than water the API gravity is
3. Bottom product of ADU is termed as
4. The temperature maintained during the thermal cracking process is
5. Define pour point
6. Aniline point is a property of.....
7. State the application of Claus process
8. Synthesis gas is mainly consisting of and
9. Name the major raw material for the production of Acrylic Acid

Part B

II. Answer any EIGHT questions from the following. Each questions carries 3 marks

8 X 3 = 24 Marks

1. Classify crude oil based on origin
2. Discuss the Crude stabilization operation
3. Define degree API
4. Prepare a block diagram of modern refinery units
5. List any six petroleum products from crude oil refining and their boiling ranges
6. Outline chemical impurities in petroleum fractions
7. Discuss the importance of Watson Characterization factor
8. List any three major Feedstock for petrochemicals and identify the products
9. List the major applications of polyethylene
10. Identify the applications of polybutadiene

Part C

Answer ALL questions. Each question carries 7 marks

6 X 7 = 42 Marks

- III. Summarize the formation of crude oil
OR
- IV. Explain the production of shale gas
- V. Summarize Visbreaking unit
OR
- VI. Discuss different types of cracking process
- VII. Explain the main elements of a hydrotreating process
OR
- VIII. Explain Catalytic polymerization processes
- IX. Summarize treatment for lube oil
OR
- X. Illustrate sweetening process of kerosene
- XI. Explain ASTM distillation curve
OR
- XII. Illustrate the measurement of TBP
- XIII. Explain production of formaldehyde from methanol
OR
- XIV. Illustrate the Manufacturing of Phenol